

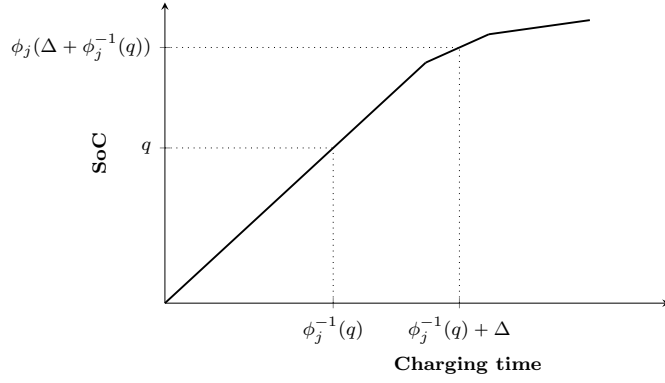


Figure 1: Piecewise linear charging function of a CS $j \in F$.

sorted in non-decreasing order of time.

Each CS $j \in F$ also has a *capacity*, given by the number of available chargers C_j . Due to the limited capacity of CSs, vehicles may incur waiting times while they queue for a charger. We therefore note that optimal solutions are not necessarily *left-shifted schedules*, as is the case for the E-VRP-NL and other E-VRP variants assuming uncapacitated CSs.

Feasible solutions to the E-VRP-NL-C must satisfy the following conditions: 1) each customer is visited exactly once by a single vehicle, 2) each route starts at the depot not before time 0 and ends at the depot not later than time T_{max} , 3) each route is energy-feasible, i.e., the SoC of an EV when it enters and leaves from any location lies between 0 and Q , and 4) no more than C_j EVs simultaneously charge at each CS $j \in F$. The objective of the E-VRP-NL-C is to minimize the total time needed to serve all customers, including driving, service, charging, and waiting times. Note that to avoid congestion at CSs, the starting times of the routes can be delayed at no cost. Moreover, an E-VRP-NL-C instance may have no feasible solution.

3 A mixed-integer linear programming formulation

We can classify explicit the mixed integer linear programming formulations for E-VRPs into two categories: CS replication-based formulations and path-based formulations. CS replication-based formulations are the compact MILP formulations typically used in the E-VRP literature to mathematically define the problem and solve toy instances. As their name suggests, in these formulations the nodes representing CSs are replicated and the number of stops at each CS replication is constrained to one. Note that in this case the number of stops at a CS is limited to its number of replications. This modeling artifice allows for tracking of the routes' travel time, distance, and SoC whenever there are several stops at the CS. To ensure that no optimal solution is cut off, the number of replications of each CS must be very large. For instance, Froger et al. (2019) provided an example where this value is equal to $4|I|$, when the congestion at each CS is neglected. CS replication-based formulations yield intractable MILPs and introduce symmetry issues.

Path-based formulations are based on a more intricate modeling strategy by which the problem is defined on a multigraph. In a nutshell, in these formulations, the nodes represent locations (the depot and the CSs or the depot and the customers) and the arcs represent possible paths between every pair of locations. A path may be defined as a sequence (possibly empty) of visits to customers between two CSs or the depot (Bruglieri et al., 2019a,b) or as a sequences of stops at CSs between two customers or the depot (Roberti and Wen, 2016; Andelmin and Bartolini, 2017; Froger et al., 2019). We refer to the former case as a *customer path* and to the latter as a *CS path*. In both cases, dominance rules are defined to limit the number of paths taken into consideration. These formulations no longer require CS vertices replication.

There also exist set partitioning-based formulations (see for instance Desaulniers et al. (2016)), but they are out of the scope of this paper. Table 1 provides details on the formulations used in the E-VRP research.

We introduce a path-based formulation for the E-VRP-NL-C. We decided to use CS paths rather than customer

paths for two reasons. First, the number of customer paths grows much faster than the number of CS paths. Second, customer path-based formulations require introducing a sufficient number of clones of paths that do not visit any customer between two CSs, whereas a CS path can at most be traveled by a single EV. We point out that the modeling of the CS capacity constraints we introduce below can easily be adapted to customer path-based formulations.

The concept of CS paths leads to a redefinition of the problem on a directed multigraph $G = (V, P)$, where $V = \{0\} \cup I$, and P is the set of arcs connecting the vertices of V . An arc in P represents a CS path p , starting in vertex $\text{org}(p)$ visiting a sequence of CSs or none and ending in vertex $\text{dest}(p)$. We denote $\mathbf{n}(p)$ the number of CSs in p . If $\mathbf{n}(p)$ is equal to 0, then p does not stop at any CS between $\text{org}(p)$ and $\text{dest}(p)$. Otherwise, we denote by $\text{cs}(p, l)$ the l -th CS visited in p (with $l \in \{1, \dots, \mathbf{n}(p)\}$). We define inputs $\mathbf{e}(p)$ and $\mathbf{t}(p)$ as the energy consumption (energy used to travel the arcs of the path) and the driving time (time to travel the arcs of the path) initially associated with CS path p . Given two vertices $i, j \in V$, we define P_{ij} as the set of CS paths connecting i to j .

Our path-based formulation of the E-VRP-NL-C involves the following decisions variables. The binary variable x_p is 1 if and only if an EV travels CS path $p \in P$. The continuous variables τ_p and y_p track the time and SoC of an EV when it departs from vertex $\text{org}(p)$ to $\text{dest}(p)$ using CS path p . To model the charging process at $\text{cs}(p, l)$, we introduce a group of variables. The continuous variables $\underline{q}_{pl}$ and $\bar{q}_{pl}$ specify the SoC of an EV when it enters and leaves $\text{cs}(p, l)$. For $k \in B_i \setminus \{0\}$, the binary variables $\underline{w}_{plk}$ and $\bar{w}_{plk}$ are equal to 1 if and only if the SoC is between $q_{\text{cs}(p, l), k-1}$ and $q_{\text{cs}(p, l), k}$ when the EV enters and leaves $\text{cs}(p, l)$. The variables $\underline{a}_{pl}$ and $\bar{a}_{pl}$ are the scaled arrival and departure times, according to the charging function of $\text{cs}(p, l)$. The continuous variables $\underline{\lambda}_{plk}$ and $\bar{\lambda}_{plk}$ represent the coefficients associated with the breakpoints $(a_{\text{cs}(p, l), k}, q_{\text{cs}(p, l), k})$ of the piecewise linear charging function, when the EV enters and leaves $\text{cs}(p, l)$. The continuous variable Δ_{pl} and ∇_{pl} represents the duration of the charging operation performed at $\text{cs}(p, l)$ and the waiting time incurred before charging at this CS. Lastly, the continuous variables $\underline{s}_{pl}$ and $\bar{s}_{pl}$ represent the starting and completion time of the charging operation performed at $\text{cs}(p, l)$.

To model the CS capacity constraints, we propose a flow-based formulation inspired from the one introduced by Artigues et al. (2003) for the Resource Constrained Scheduling Problem. For every CS $j \in F$, we consider C_j resources (represented by the chargers). Each resource can execute at most one operation at any given time. For convenience, we introduce the set O_j of potential charging operations at CS j . Every operation $o \in O_j$ represents a stop at j at a specific position $l(o)$ in a CS path going from $\text{org}(o)$ to $\text{dest}(o)$. Although there may exist several CS paths between two specific vertices that stop at j at the same position, at most one these CS paths can be selected in a solution. We also introduce two dummy operations $\mathbf{o}_j^+$ and $\mathbf{o}_j^-$, which act as the source and the sink of the flow for each CS j . Specifically, the resources flow from $\mathbf{o}_j^+$ to $\mathbf{o}_j^-$ through the charging operations. Let $(o, o') \in (O_j \cup \{\mathbf{o}_j^+\}) \times (O_j \cup \{\mathbf{o}_j^-\})$ be a couple of charging operations; then the continuous flow variable $f_{oo'}$ denotes the number of chargers that are transferred from operation o to operation o' . When both these operations are not dummy, $f_{oo'}$ is equal to one if and only if these operations are scheduled on the same charger, o' is scheduled after o , and no other operation is scheduled on the charger between the completion of o and the beginning of o' . Binary variable $u_{oo'}$ is equal to 1 if and only if operation o' starts after the completion of operation $o \neq o'$. For notational convenience, we define $\mathbf{C}(o) := 1$ for all $o \in O_j$ and $\mathbf{C}(\mathbf{o}_j^+) := \mathbf{C}(\mathbf{o}_j^-) := C_j$.

A path-based formulation for the E-VRP-NL-C, denoted as $[P^{\text{path}}]$, is as follows:

$$[F^{\text{path}}] \quad \text{minimize} \quad \sum_{p \in P} \left(\mathbf{t}(p)x_p + \sum_{l=1}^{\mathbf{n}(p)} (\Delta_{pl} + \nabla_{pl}) \right) \quad (1)$$

$$\text{subject to} \quad \sum_{j \in V, i \neq j} \sum_{p \in P_{ij}} x_p = 1 \quad i \in I \quad (2)$$

$$\sum_{j \in V, i \neq j} \sum_{p \in P_{ji}} x_p - \sum_{j \in V, i \neq j} \sum_{p \in P_{ij}} x_p = 0 \quad i \in V \quad (3)$$

$$\sum_{j \in V, j \neq i} \sum_{p \in P_{ji}} \left(y_p - \mathbf{e}(p)x_p + \sum_{l=1}^{\mathbf{n}(p)} (\bar{q}_{pl} - \underline{q}_{pl}) \right) = \sum_{j \in V, j \neq i} \sum_{p \in P_{ij}} y_p \quad i \in I \quad (4)$$

$$y_p - e_{\text{org}(p), \text{cs}(p, 1)} x_p = \underline{q}_{p1} \quad p \in P \quad (5)$$

$$\bar{q}_{p, l-1} - e_{\text{cs}(p, l-1), \text{cs}(p, l)} x_p = \underline{q}_{pl} \quad p \in P, l \in \{2, \dots, \mathbf{n}(p)\} \quad (6)$$

$$y_p - \mathbf{e}(p)x_p + \sum_{l=1}^{\mathbf{n}(p)} (\bar{q}_{pl} - \underline{q}_{pl}) \geq 0 \quad i \in I, p \in P_{i0} \quad (7)$$

$$y_p = Qx_p \quad i \in V \setminus \{0\}, p \in P_{0i} \quad (8)$$

$$y_p \leq Qx_p \quad p \in P \quad (9)$$

$$\underline{q}_{pl} = \sum_{k \in B_{\text{cs}(p,l)}} \underline{\lambda}_{plk} q_{\text{cs}(p,l),k} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (10)$$

$$\underline{a}_{pl} = \sum_{k \in B_{\text{cs}(p,l)}} \underline{\lambda}_{plk} a_{\text{cs}(p,l),k} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (11)$$

$$\sum_{k \in B_{\text{cs}(p,l)}} \underline{\lambda}_{plk} = \sum_{k \in B_{\text{cs}(p,l)} \setminus \{0\}} \underline{w}_{plk} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (12)$$

$$\sum_{k \in B_{\text{cs}(p,l)} \setminus \{0\}} \underline{w}_{plk} = x_p \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (13)$$

$$\underline{\lambda}_{pl0} \leq \underline{w}_{pl1} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (14)$$

$$\underline{\lambda}_{plk} \leq \underline{w}_{plk} + \underline{w}_{pl,k+1} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\}, k \in B_{\text{cs}(p,l)} \setminus \{0, b_{\text{cs}(p,l)}\} \quad (15)$$

$$\underline{\lambda}_{plb_{\text{cs}(p,l)}} \leq \underline{w}_{plb_{\text{cs}(p,l)}} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (16)$$

$$\bar{q}_{pl} = \sum_{k \in B_{\text{cs}(p,l)}} \bar{\lambda}_{plk} q_{\text{cs}(p,l),k} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (17)$$

$$\bar{a}_{pl} = \sum_{k \in B_{\text{cs}(p,l)}} \bar{\lambda}_{plk} a_{\text{cs}(p,l),k} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (18)$$

$$\sum_{k \in B_{\text{cs}(p,l)}} \bar{\lambda}_{plk} = \sum_{k \in B_{\text{cs}(p,l)} \setminus \{0\}} \bar{w}_{plk} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (19)$$

$$\sum_{k \in B_{\text{cs}(p,l)} \setminus \{0\}} \bar{w}_{plk} = x_p \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (20)$$

$$\bar{\lambda}_{i0} \leq \bar{w}_{pl1} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (21)$$

$$\bar{\lambda}_{plk} \leq \bar{w}_{plk} + \bar{w}_{pl,k+1} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\}, k \in B_{\text{cs}(p,l)} \setminus \{0, b_{\text{cs}(p,l)}\} \quad (22)$$

$$\bar{\lambda}_{plb_{\text{cs}(p,l)}} \leq \bar{w}_{plb_{\text{cs}(p,l)}} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (23)$$

$$\Delta_{pl} = \bar{a}_{pl} - \underline{a}_{pl} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (24)$$

$$\sum_{j \in V, j \neq i} \sum_{p \in P_{ji}} \left(\tau_p + \mathbf{t}(p)x_p + \sum_{l=1}^{\mathbf{n}(p)} (\Delta_{pl} + \nabla_{pl}) \right) + g_i = \sum_{j \in V, j \neq i} \sum_{p \in P_{ij}} \tau_p \quad i \in I \quad (25)$$

$$\tau_p + \sum_{l=1}^{\mathbf{n}(p)} (\Delta_{pl} + \nabla_{pl}) \leq (T_{\max} - \mathbf{t}(p) - g_{\text{dest}(p)} - t_{\text{dest}(p),0}) x_p \quad p \in P \quad (26)$$

$$\tau_p + t_{\text{org}(p), \text{cs}(p,1)} x_p + \nabla_{p1} = s_{p1} \quad p \in P \quad (27)$$

$$\bar{s}_{p,l-1} + t_{\text{cs}(p,l-1), \text{cs}(p,l)} x_p + \nabla_{pl} = s_{pl} \quad p \in P, l \in \{2, \dots, \mathbf{n}(p)\} \quad (28)$$

$$\bar{s}_{pl} = s_{pl} + \Delta_{pl} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (29)$$

$$\sum_{o' \in O_j \cup \{o_j^+\}} f_{o'o} = \sum_{\substack{p \in P_{\text{org}(o), \text{dest}(o)} \\ : \text{cs}(p, l(o)) = j(o)}} x_p \quad j \in F, o \in O_j \quad (30)$$

$$\sum_{o' \in O_j \cup \{o_j^+\}} f_{o'o} - \sum_{o' \in O_j \cup \{o_j^-\}} f_{oo'} = 0 \quad j \in F, o \in O_j \quad (31)$$

$$\sum_{o \in O_j \cup \{o_j^-\}} f_{o_j^+, o} = C_j \quad j \in F \quad (32)$$

$$\sum_{o \in O_j \cup \{o_j^+\}} f_{o, o_j^-} = C_j \quad j \in F \quad (33)$$

$$\sum_{\substack{p \in P_{\text{org}(o), \text{dest}(o)} \\ : \text{cs}(p, l(o)) = j(o)}} s_{p, l(o)} - \sum_{\substack{p \in P_{\text{org}(o'), \text{dest}(o')} \\ : \text{cs}(p, l(o')) = j(o')}} \bar{s}_{p, l(o')} \geq T_{\max} (u_{o'o} - 1) \quad j \in F, o, o' \in O_j \quad (34)$$

$$f_{oo'} \leq \min(\mathcal{C}(o), \mathcal{C}(o')) u_{oo'} \quad j \in F, (o, o') \in (O_j \cup \{o_j^+\}) \times (O_j \cup \{o_j^-\}) \quad (35)$$

$$x_p \in \{0, 1\} \quad p \in P \quad (36)$$

$$\tau_p \geq 0, y_p \geq 0 \quad p \in P \quad (37)$$

$$\underline{q}_{pl}, \bar{q}_{pl}, \underline{a}_{pl}, \bar{a}_{pl}, \underline{s}_{pl}, \bar{s}_{pl}, \Delta_{pl}, \nabla_{pl} \geq 0 \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\} \quad (38)$$

$$\underline{\lambda}_{plk} \geq 0, \bar{\lambda}_{plk} \geq 0 \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\}, k \in B_{cs(p,l)} \setminus \{0\} \quad (39)$$

$$\underline{w}_{plk} \in \{0, 1\}, \bar{w}_{plk} \in \{0, 1\} \quad p \in P, l \in \{1, \dots, \mathbf{n}(p)\}, k \in B_{cs(p,l)} \setminus \{0\} \quad (40)$$

$$u_{oo'} \in \{0, 1\} \quad j \in F, o, o' \in O_j \quad (41)$$

$$f_{oo'} \geq 0 \quad j \in F, (o, o') \in (O_j \cup \{o_j^+\}) \times (O_j \cup \{o_j^-\}) \quad (42)$$

The objective function (1) minimizes the total driving, charging, and waiting time. Constraints (2) ensure that each customer is visited exactly once. Constraints (3) impose flow conservation: at each vertex the number of incoming EVs is equal to the number of outgoing EVs. Constraints (4) track the SoC of EVs at each customer location. Constraints (5) track the SoC at the arrival at the first CS of each CS path. Constraints (6) couple the SoC of an EV that leaves a CS with its SoC upon arrival to the next CS of the same path. Constraints (7) ensure that if the EV travels from a customer to the depot, it has sufficient energy to reach its destination. Constraints (8) state that every EV leaves the depot with a fully charged battery. Constraints (9) couple the SoC tracking variables with the path travel variables. Constraints (10)–(24) model the relationship between the SoCs of an EV upon entering and leaving a CS and the charging time. Constraints (25) track the departure time at each customer. Constraints (26) couple the time tracking variable with the path travel variables, and impose the route duration limit. Constraints (27) and (28) define the starting and completion times of every charging operation, as well as the potential waiting time before the start of the operation. Constraints (29) define the completion time of the charging operations based on their starting time and duration. Constraints (30) state that a charger has to be allocated to every charging operation of each selected CS path (at most one CS path in the sum can be selected in a solution). Constraints (31) ensure flow conservation of the chargers of each CS. Constraints (32) and (33) compute the flow value at the beginning and at the end of the time horizon, which is equal to the number of available chargers at each CS. Constraints (34) couple the sequencing variables with the starting time of the corresponding charging operations. Constraints (35) couple the flow variables with the operation sequence variables. Specifically, a charger can be sent from a charging operation o' to another charging operation o if o starts after the completion of o' . Finally, constraints (36)–(42) set the domains of the decision variables.

Without preprocessing, the number of CS paths explodes with the number of CSs and the number of customers. However, a large number of these arcs cannot be part of an optimal solution. Froger et al. (2019) presented a filtering procedure to reduce the number of CS paths based on the definition of a dominance rule between two CS paths having the same origin and destination. Due to the potential waiting times that can occur at CSs, we cannot apply the dominance rule described in the work of Froger et al. (2019) between CS paths with the same origin and destination if they both contain CSs. This is due to the fact that waiting times depend on all the CS paths selected to build a complete solution. In our computational experiments, we apply the dominance rule between the unique CS path with no CS between i and j (if it exists) and every other CS path of P_{ij} , with i and j in V .

4 Solution method

In this section, we introduce a matheuristic to solve the E-VRP-NL-C. The method relies on an algorithmic framework based on two interacting components: a route generator and a solution assembler. The first component builds a pool of high-quality and diverse routes exploring the solution space of the problem without CS capacity constraints, while the second component recombines routes from the pool trying to build a solution satisfying the CS capacity constraints.

Algorithm 1 outlines the general structure of the method. The algorithm starts by generating an initial solution without taking the CS capacity constraints into account (line 1). In our implementation, this step is carried out using a modified version of the classical Clarke and Wright heuristic. Next, the algorithm enters the main loop (lines 3–27). During n_{max} iterations the algorithm alternates between the *route generation* and *solution assembly* phases. In particular, the algorithm uses procedure **generateRoutes**($\cdot$) to retrieve a triplet $(\Phi_O, \Phi_R, \Omega^{ST})$, where Φ_O, Φ_R are sets of high-quality solutions for the *original* and *relaxed* problems (i.e., with and without CS capacity constraints), and Ω^{ST}